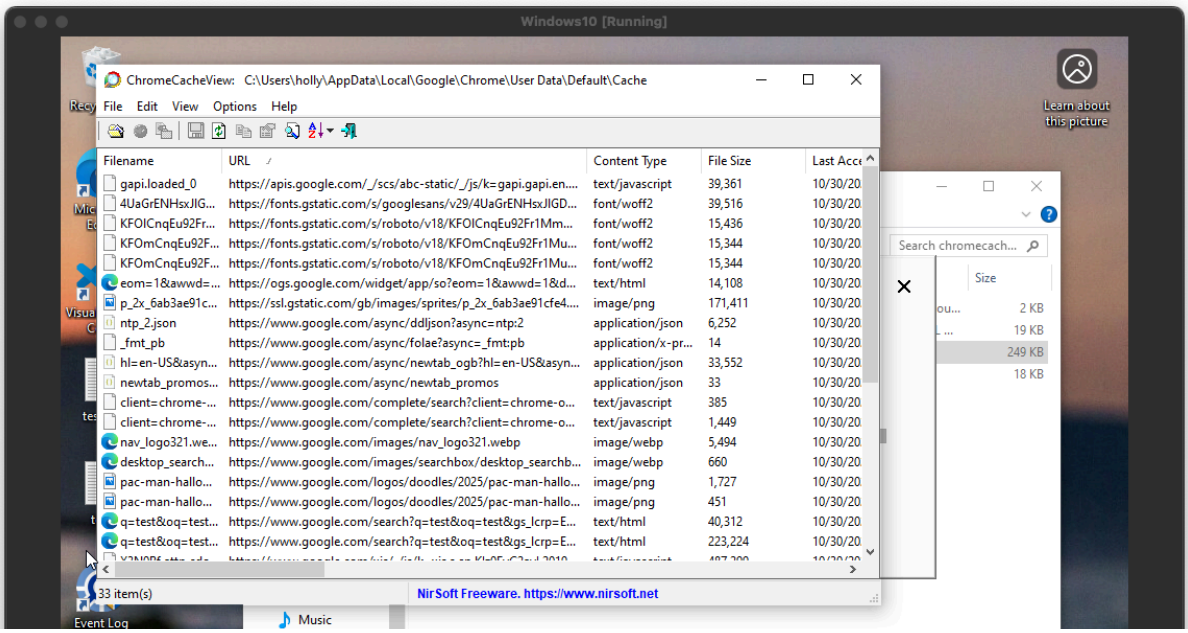
Holly Pasquinelli
Lab 5

Windows Forensics Part II

Non-technical Summary

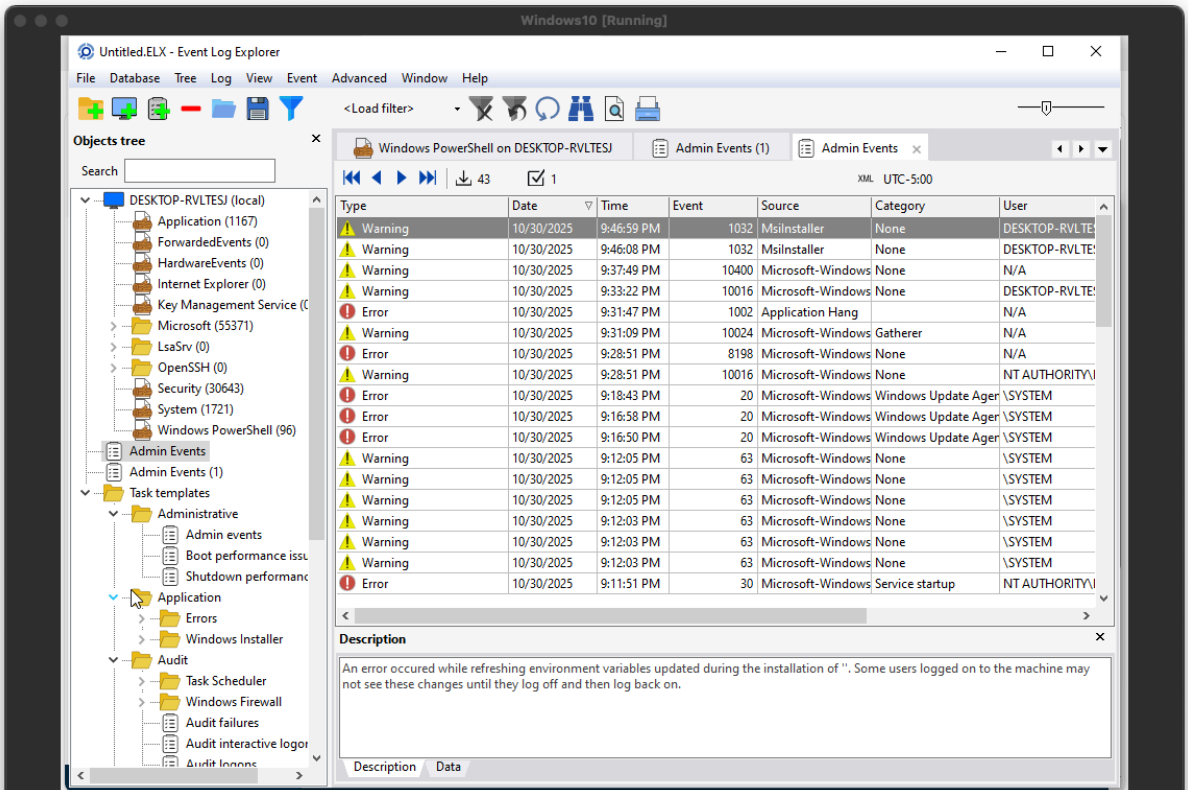
In this lab, using tools such as NIRSOFT, OSForensics, and Python, we are able to access browser artifacts such as cache, cookies, and history. We also explored how to monitor and understand windows events, and handle windows registry, recycle bin, and browser artifacts with Python. In regards to digital forensics, these tools help investigators understand what has happened on a computer, and how applications have been used. The ability to access information such as registry and recycle bin allow us to recover files using winreg and winshell libraries. Additionally, we can use the os library to access browser cache, history and cookies, which helps retrace user activity. These logs give us information about downloads, websites accessed, and activity. User activity can give investigators an idea of what the user's intentions are, what they may be trying to access, as well as their motive. Windows events give us supplementary information and can provide us with specific event times, suspicious events, and a variety of logs such as system and security logs.

Technical Report

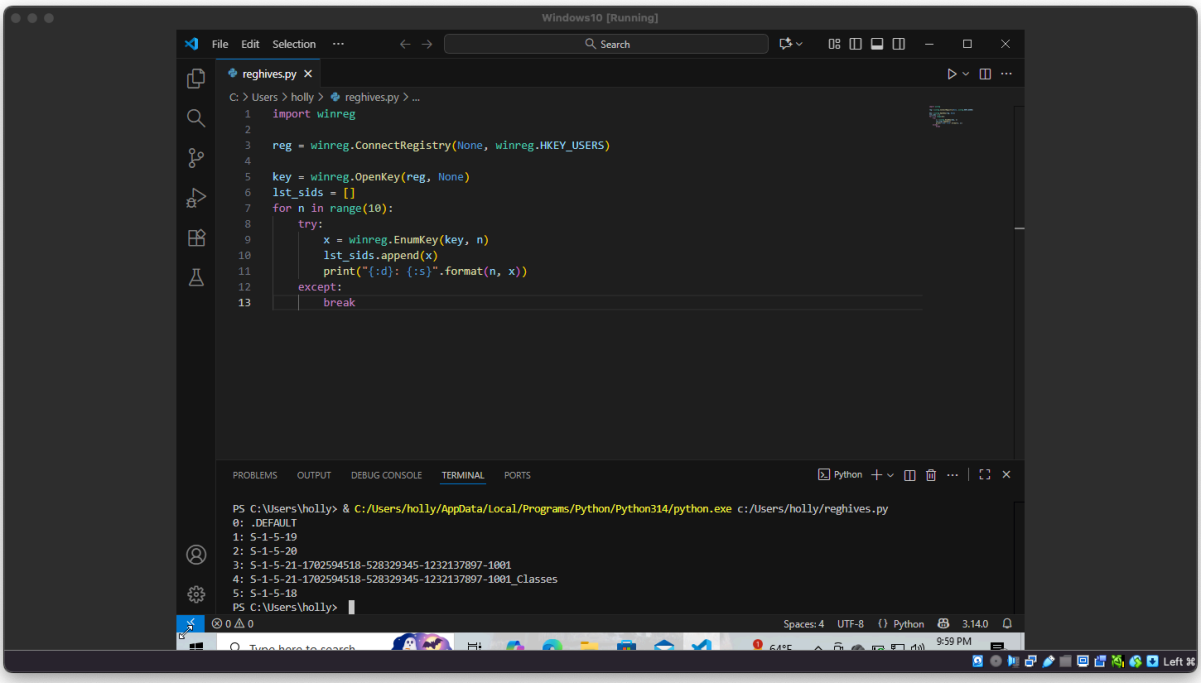


For starters we examined information from web browsers. Browsers contain information such as caches, cookies, and history that can paint a picture of user activity. In the lab, we used ChromeCacheView to display cache data. As pictured above, we

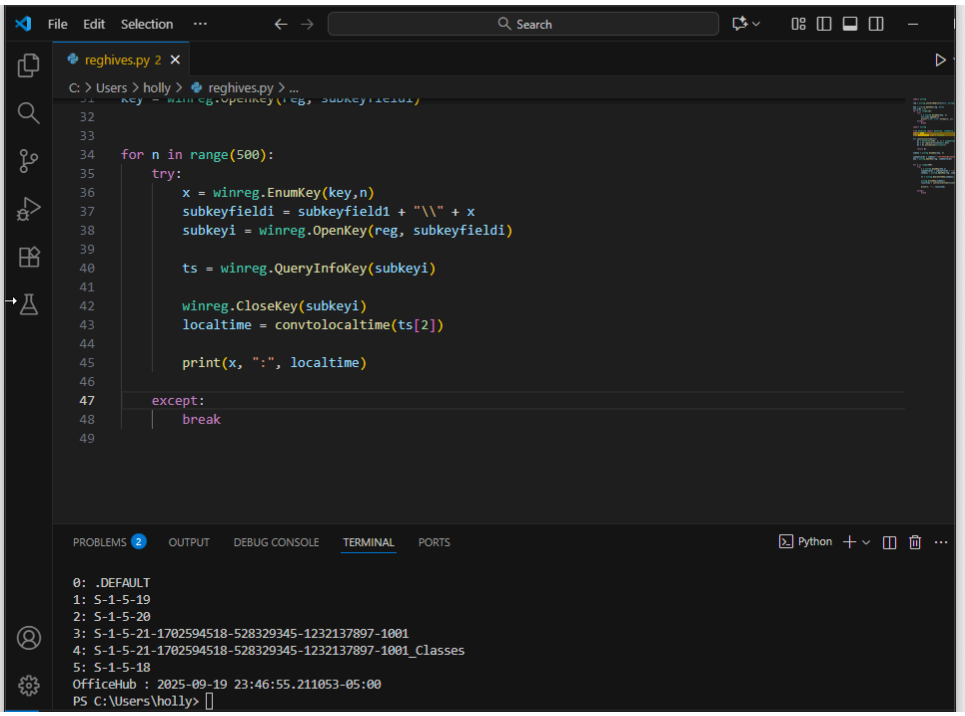
can see that this displays the filename, URL of the website, content type such as image, text, application, filesize, and date last accessed. We also used ChromeHistoryView to display history data. This displays information such as the URL of the site, title, when the site was visited, the number of times the site was visited, how long it was visited for, visit ID, profile, transition (typed, link, form), and history file. Lastly we used ChromeCookiesView to display cookie data. This displays information such as the host site, path, name, value, if it's secure or http only, date and time last accessed, creation date and time, and expiration date and time.



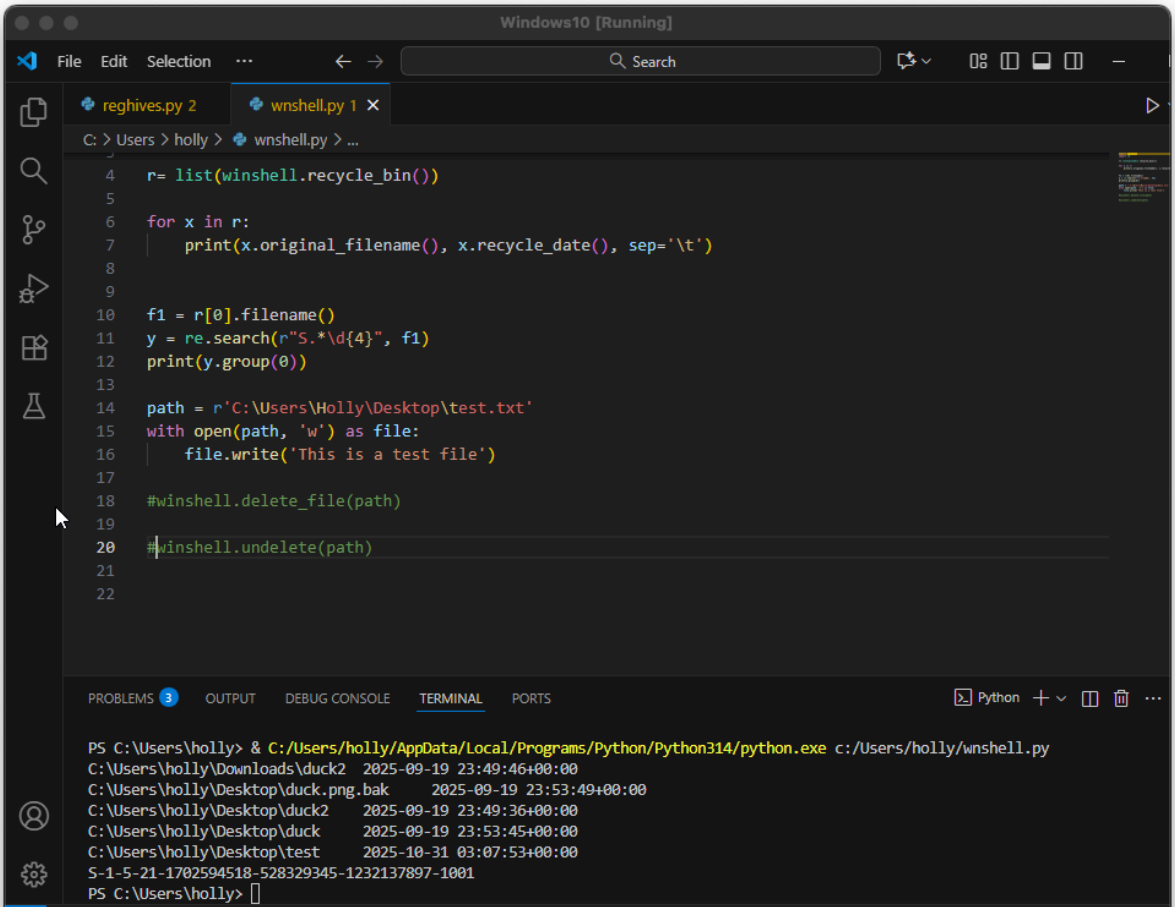
Next, we looked into accessing windows events logs. On Windows 11, we looked at the log files in the folder C:\Windows\System32\config[.]. Using Event Log Explorer, I looked at the log files under DESKTOP-RVLTESJ, the name of the windows machine I was using. Here, under admin events we can see the type of event, warning, or error, the date and time of the event, event ID, source, category, and user. For example, we can see there are 3 errors in the middle of the list for Microsoft-Windows where a windows update was attempted by the system. We can see by the type of event that this update failed 3 times. This is an important event because if a system fails to update, it can leave vulnerabilities and opportunities for unauthorized users to gain access to a system. This allows investigators to see where a potential vulnerability might have occurred, or when it does occur, allow a chance to troubleshoot before damage is done.



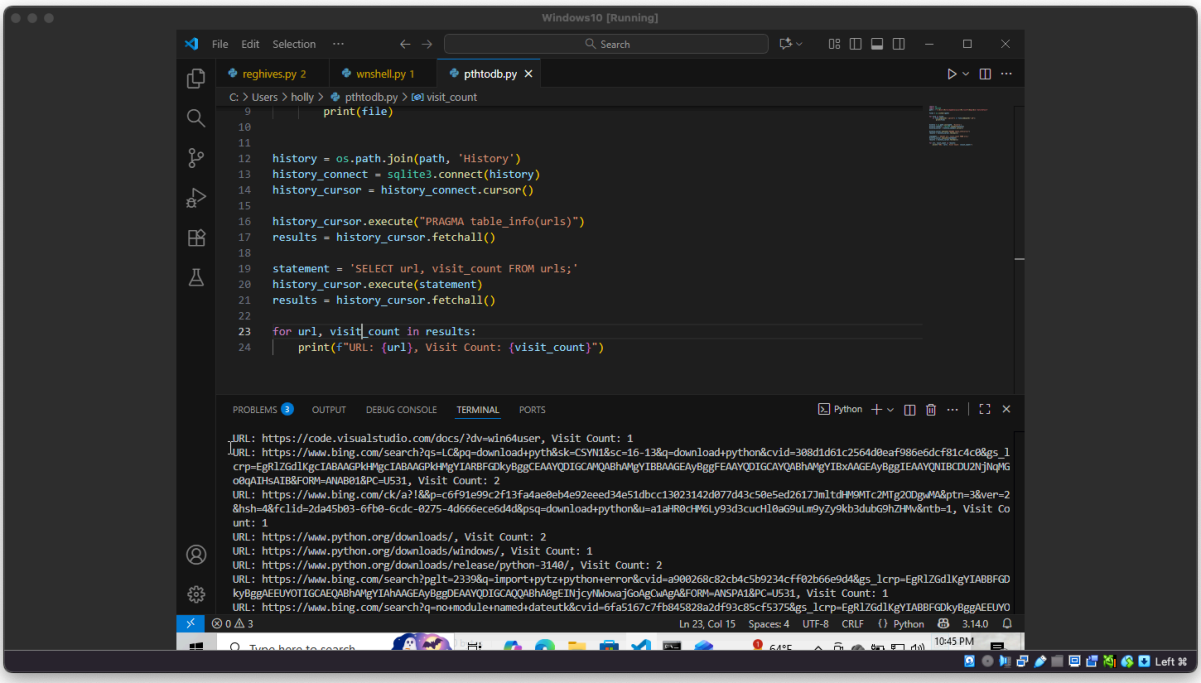
Next, using Python, we used the winreg library to gather information about the windows registry. We used the following to display information about the content of HKEY_USERS hive. We can see the output returns enumerated subkeys and their SIDs. If we look at 3 and 4, we can see that 1001 refers to the first user created upon system creation.



Next, we accessed registry keys and values. Here, we specified the Microsoft Office Key. We can see this returned the OfficeHub subkey under HKEY_USERS with the date and time last modified. For this virtual machine, I do not have Office installed, otherwise we would see other subkeys such as Outlook, PowerPoint, Word, etc displayed along with their subkey, number of values, and time last modified.



Next, we used the winshell library to access and handle the recycle bin. We can see the output is deleted files, names, and date that they were deleted and entered the recycle bin. Using the code above, we store the user SID who deleted the file, created another file test.txt, delete, and undelete the file. This is especially important in digital forensics because it can help investigators find and trace deleted files, and activity on files. This also gives us information about when a file was last modified.



Lastly, we used Python to display browser history, cookies, and caches. In the lab, we looked at the Mozilla Firefox Web Browser. On my windows virtual machine, I had Microsoft Edge installed so I decided to try and generalize the code to adapt to the Edge browser. I discovered that the Edge browser also stores .sqlite database files. We use the os library to get the content of the path folder. Above, the path specified is r'C:\Users\Holly\AppData\Local\Microsoft\Edge\User Data\Default. For Firefox, to display the files, we also use .endswith(".sqlite") and .endswith(".db"). Next, we access the browsing history. Instead of 'places.sqlite' we use 'History' to store the history. To display the table's information for Edge, we use the table name urls instead of moz_places. To display results, I decided to print the URL, and visit count. Of course, the output is quite messy. As we can see above, we can see that I recently downloaded Python, for the purpose of this lab. We can see that I visited the site for downloads twice, and specific for windows once.

In regards to digital forensics, we can use this to understand user activity. If we see a high visit count for a certain site, it may suggest that an unauthorized user is attempting a brute force attack to attempt to gain access to something. It also establishes a user's pattern, and behaviors. This can help identify activity that is out of the ordinary, and prompt further investigation into a certain time frame, or site that is being accessed.

References

1. Lab 8 pdf
2. winreg Python documentation https://docs.python.org/3/library/winreg.html